浙江大学 2023–2024 学年秋冬学期

《计算机组成与设计》课程期末考试试卷

课程号：111C0130，开课学院：信息与电子工程学院

考试试卷：√A 卷、B 卷（请在选定项上打√）

考试形式：√闭、开卷（请在选定项上打√），

允许带 1 张 A4 纸大小的手写资料和计算器入场

考试日期：2024 年 1 月 16 日，考试时间：120 分钟

诚信考试，沉着应考，杜绝违纪。

考生姓名：_____学号：_____所属院系（专业）：_____

题序	一	二	三	四	五	总 分
得分						
评卷人						

I. CHOICE (Only one correct answer, 60 points)

1.The DRAMs are basically used as _____ ( ) C

A. register

B. cache

C. main memory

D. disk

2. Consider the following C code:

```
typedef unsigned char *pointer; // sizeof(unsigned char) = 1 byte

void show_bytes(pointer start, size_t len) {
   for (int i = 0; i < len; i++)
        printf("0x%x\n", start[i]);
}

int main() {
   int a = 0x44332211;
   show_bytes((pointer) &a, sizeof(int));
}
```

If this C code runs on a little-endian machine, what will we get on the **third** line of the terminal output? () C

A. 0x11

B. 0x22

C. 0x33

D. 0x44

3.The size of the register in the RISC-V architecture is _____()C

A. 8 bits

B. 16 bits

C. 32 bits

D. 64 bits

4. For a given function, which programming language likely takes the most lines of code?()C

A. C

B. Java

C. RISC-V assembly language

D. Python

5. What is the decimal value of this 32-bit two's complement number?()B

$1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1111\ 1000_{two}$

A. $4\ 294\ 967\ 288_{ten}$

B. -8_{ten}

C. -4_{ten}

D. -2_{ten}

6. Which operations can isolate a field in a word?()A

A. A shift left followed by a shift right

B. AND

C. ADD

D. SUB

7. Which one of the following instructions is used for making decisions?()D

A. add

B. or

C. and

D. beq

8. Which of the following statements must be true about compilers?()A

A. Compiled code generally is only able to run on one ISA.

B. The code produced is always more efficient and higher performance than that produced by

interpreters.

C. There is only one compiler per language.

D. Compilers are always more difficult to write than interpreters.

9. In a system which has 128 registers, the register id is _____ long.()C

A. 32-bit B. 8-bit C. 7-bit D. 6-bit

10. The execution of the following two instructions may have the _____.()A

add x5,x3,x4

sd x5,0(x6)

A. RAW (Read after Write) B. WAW (Write after Write)

C. WAR (Write after Read) D. No data dependency

11. The temporal aspect of the locality of reference means (_____). C

A. That the recently executed instruction won't be executed soon

B. That the recently executed instruction is temporarily not referenced

C. That the recently executed instruction will be executed soon again

D. None of the mentioned

12. Given the following RISC-V assembly code (and assuming all registers start at 0):

```
        addi    t1, x0, 10
        add     t2, t1, t1
repeat: addi    t2, t2, -4
        add     t3, t2, t2
        addi    t1, t1, -2
        bne     x0, t1, repeat
```

What is the final value of register t3? A

A.0

B.-4

C.8

D.4

index:1bits,offset:3bits block:4

13. You have a two-way set-associative cache with 8B blocks and a total size of 32B. Suppose use a least-recently used replacement policy and begin from power on. Given this sequence of byte addressed accesses, what is the hit rate? (_____) C

0, 4, 128, 24, 224, 88, 134, 92

A. 12.5%

B. 25.0%

C. 37.5%

D. 50.0%

tag=32-11-4=32-15=17

index:11

entry:2K

offset: 4bits

block size:16 bytes

block:4K

14. For a 32-bit cache-memory system, a 64KB, 2-way set-associative cache has 4 words cache line size, how many bits are there in such cache's tag?()A

A. 17 B. 19 C. 21 D. 23

15. The reason for the implementation of the cache memory is _____. ()B

A. to increase the internal memory of the system

B. the difference in speeds of operation of the processor and memory

C. to reduce the memory access time

D. all of the mentioned

16. A page fault occurs when _____. C

A. there is an error in a specific page

B. a program accesses a page of main memory

C. a program accesses a page not currently in main memory

D. a program accesses a page belonging to another program

17. When a subroutine is called, the address of the instruction following the CALL instruction stored in/on the__. D

A. stack pointer

B. accumulator

C. program counter

D. stack

entry=2^10

index:32-3-19=32-22=10bits

block:4K=2^12 offset:3bits

18. For a 32-bit cache-memory system, a 32 KB cache has 8 bytes cache line size. If the cache's tag needs 19 bits, the cache is _____. C

set=block/entry=4

A. direct-mapped

B. 2-way set-associative

C. 4-way set- associative

D. full-associative.

19. Assum the initial value in x10 is 1910. What is the value of x10 for the following sequence of RISC-V instructions? C

slli x10, x10, 2 *4

xori x10, x10, 0xfff

A. 76 B.-76 C.-77 D.179

20. Let A be a 1024×1024 matrix of 32-bit int elements stored in row-major order, aligned to the beginning of a cache line.

```
for (int i = 1; i < 16; i++) {
int x = A[0][i-1];
int y = A[i][i];
A[i][i] = x + y;
}
```

Assume that memory accesses are executed in the order shown in the program – i.e., the compiler does not reorder load and store instructions. Variables x, y, and i are held in registers.

block:256 offset:4bits

Consider a 4 KiB direct-mapped L1 data cache with 16-byte cache lines. Assume that the cache is initially empty. The numbers of cache hits on the loop shown above is ___. B

A.16
B.15
C.14
D.0

II. TRUE OR FALSE (10 points)

1.___ The LRU can be improved by providing a little randomness in the access. T

2.___ The direct mapping is costlier than associative mapping. F

3.___ In the memory hierarchy, as the speed of operation increases the memory size also increases. F

4.___ Virtual memory allows a single program to expand its address space beyond the limits of main memory. T

5.___ A write-through cache typically requires less bus bandwidth than a write-back cache. F

6.___ An executable binary file that can run on a RISC-V CPU may not be able to execute on another RISC-V CPU. T

7.___ The CPI of superscalar processors can be less than one. T

8.___ The page table is store in the main memory. T

9.___ Cache block size (B) can only affect miss latency. F

10.___ There is no way to reduce compulsory misses. F

III. Assembly and Pipeline (10 points)

Consider the following sequence of instructions:

```
add   x15, x12, x11
lw    x13, 8(x15)   2
lw    x12, 0(x2)    1
or    x13, x15, x13
```

2
sw x13, 0(x15)

Assume each datapath stage requires the following amount of time to complete:

- Instruction fetch (IF): 30 ns
- Instruction decode (ID): 20 ns
- Execute / address calculation (EX): 25 ns
- Memory access (MEM): 30 ns
- Register write back (WB): 20 ns

a) How long will this sequence of instructions take in a single-cycle datapath? ___625____ns **(2 points)**
b) If we assume a pipelined datapath without forwarding, how long will one loop iteration take in a pipelined datapath? ___420____ns **(2 points)**
c) If we now assume a pipelined datapath with forwarding, how many data hazard(s) cannot be solved with forwarding? (number only)
__0____ **(3 points)**
d) If we now assume a pipelined datapath with forwarding, how long will one iteration take?
__270_____ns **(3 points)**

IV. CACHE (10 points)

Given the following looping workload over an array where N is a large power of 2. The cache starts out empty, and the process() function doesn't introduce any significant cache pressure (so you can discount any hits or misses in the process() function)

```
uint32_t arr[N];

for (int j=0; j < 30; j++) {
    for (int i=0; i < N; i += 1) {
        process(arr[i]);
    }
}
```

Express the following answers as a function of N.

block=N

1. Suppose we have a LRU fully associative cache of size 4N B and a block size of 4B.
   Number of hits: _____ (2 points) 29N
   Number of misses: _____ (2 points) N

block=N/2

2. Suppose we have a LRU fully associative cache of size 2N B and a block size of 4B:
   Number of hits: _____ (2 points) 0
   Number of misses: _____ (2 points) 30N

block=N/4

3. Suppose we have a LRU 2-way set-associative cache of size 2N B with a block size of of 8B:

Number of hits: _____ (2 points) 15N

V. VIRTUAL MEMORY (10 points)

We are given a system with physical memory of size 4 GiB 2^32 and virtual memory of 16 GiB 2^34 and a page size of 8 KiB 2^13. Our page table system has one level with no TLB.

1. How many bits are in the page offset? _____ (2 points) 13
2. How many bits are in the VPN? _____ (3 points) 21 Ventry=2^21
3. How many bits are in the physical address? _____ (2 points) 32
4. How many bits are in the PPN? _____ (3 points) 19 Pentry=2^19